

The background of the slide is a dark blue field filled with a complex, glowing network of thin lines and small dots, resembling a data visualization or a molecular structure. The lines and dots are in various shades of blue and cyan, creating a sense of depth and connectivity.

Introduction to Big Data

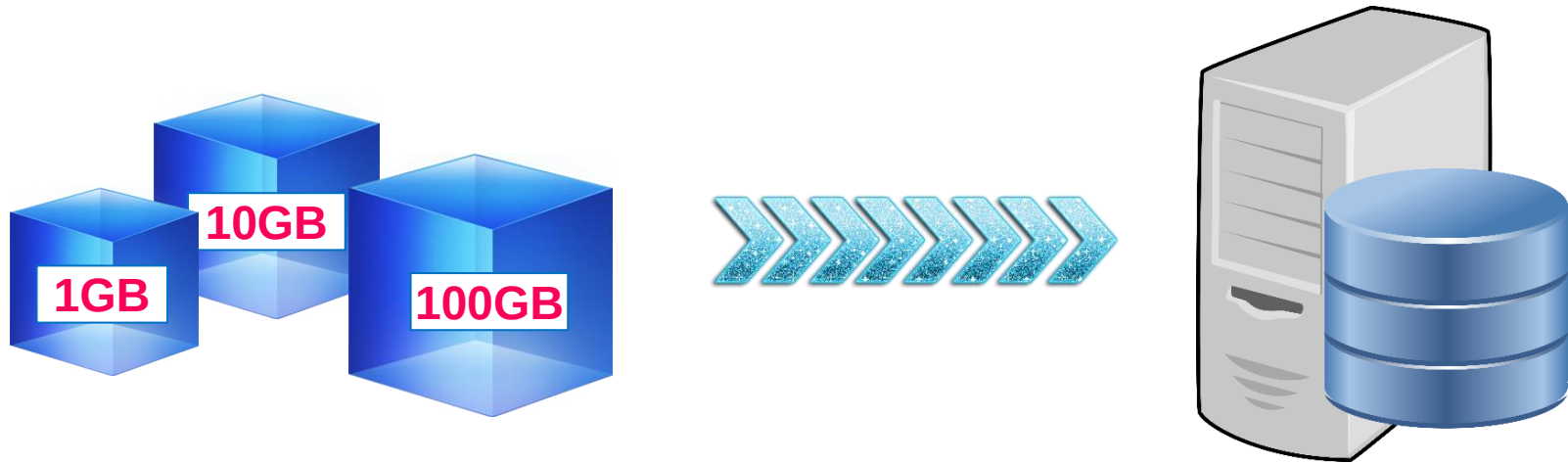
APACHE HADOOP

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Outline

- An introduction to Apache Hadoop
- When to use and not to use Hadoop
- Hadoop installation
- Hadoop: Now and Future

Imagine a scenario where...



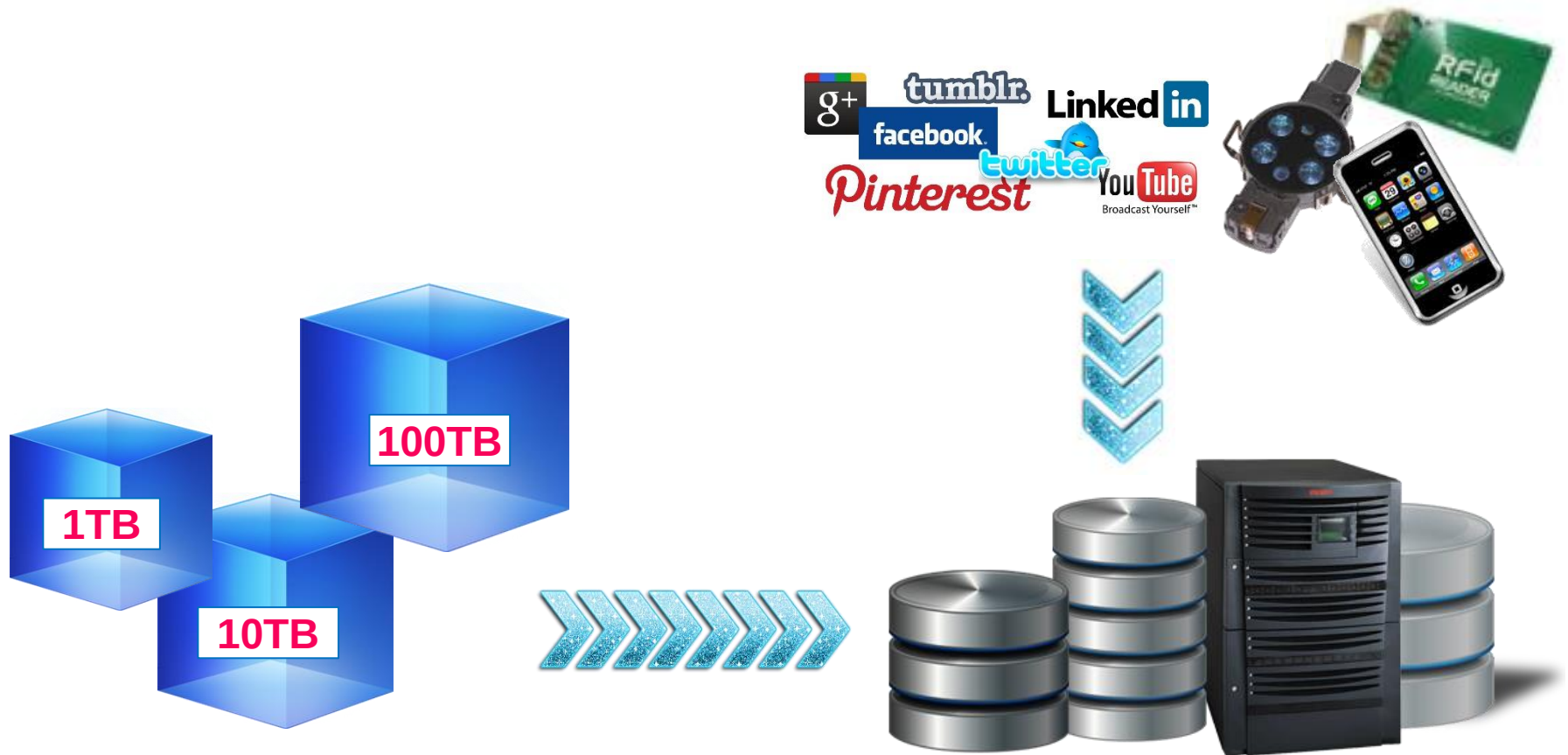
You have 1GB of data to process.... No problem!

Your company starts growing very quickly, and that data arises to 10GB

And then 100GB....

You start to reach the limits of your current desktop computer.

Imagine a scenario where...



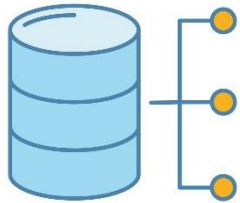
Therefore, you scale-up by investing in a larger computer, but...

the data increases quickly in a few months

it is required to feed the application with unstructured data

Imagine a scenario where...

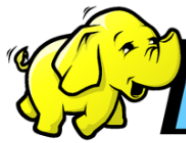
You may want to



Derive information from data of various types, not limited to relation data only.



Obtain the derivation as soon as possible.



hadoop may be your answer!!!

What is Hadoop?



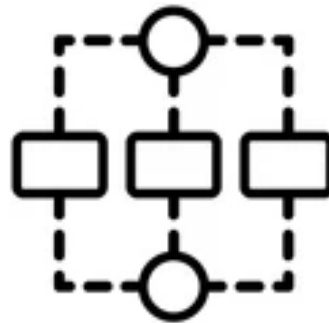
Written in **Java**



ASF project



Distributed storage



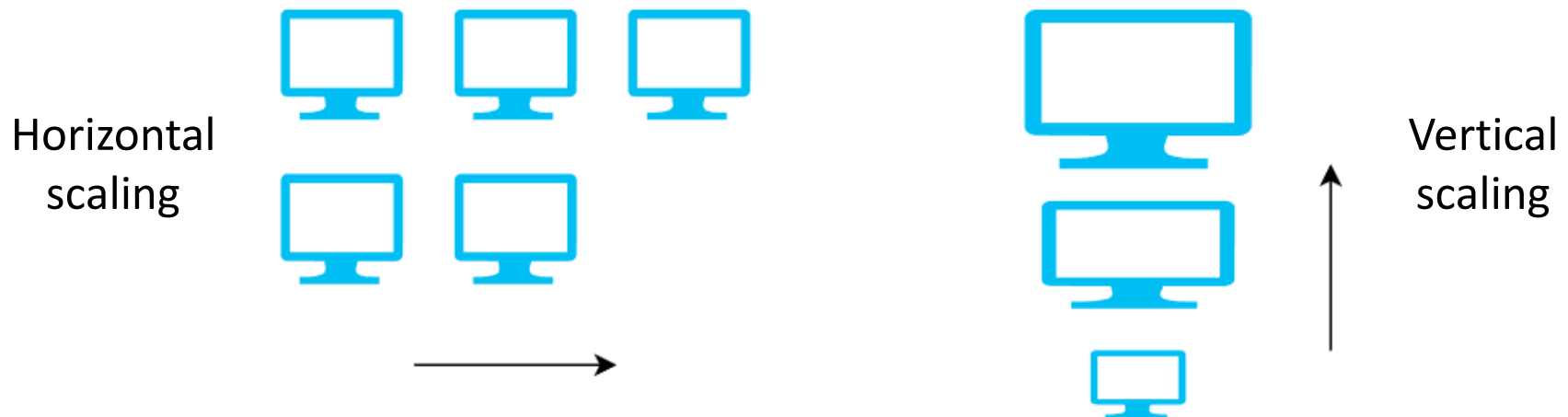
Massive parallel
processing



Reliable replication

Horizontal vs. Vertical scaling

- **Horizontal scaling** (Scaling out): Add **more machines or nodes** to a system to **distribute the load**.
 - E.g, expand a Hadoop cluster by adding more nodes.
- **Vertical scaling** (Scaling up): **Increase the capacity of a single machine** by adding more resources (CPU, RAM, storage).
 - E.g, upgrade a server from 16GB RAM to 64GB RAM



Horizontal scaling: Pros and Cons

Advantages

- More fault-tolerant (if one node fails, others can take over)
- Can handle massive scalability needs
- Cost-effective for cloud-based infrastructure.

Disadvantages

- Requires load balancing and data distribution.
- More complex system management.

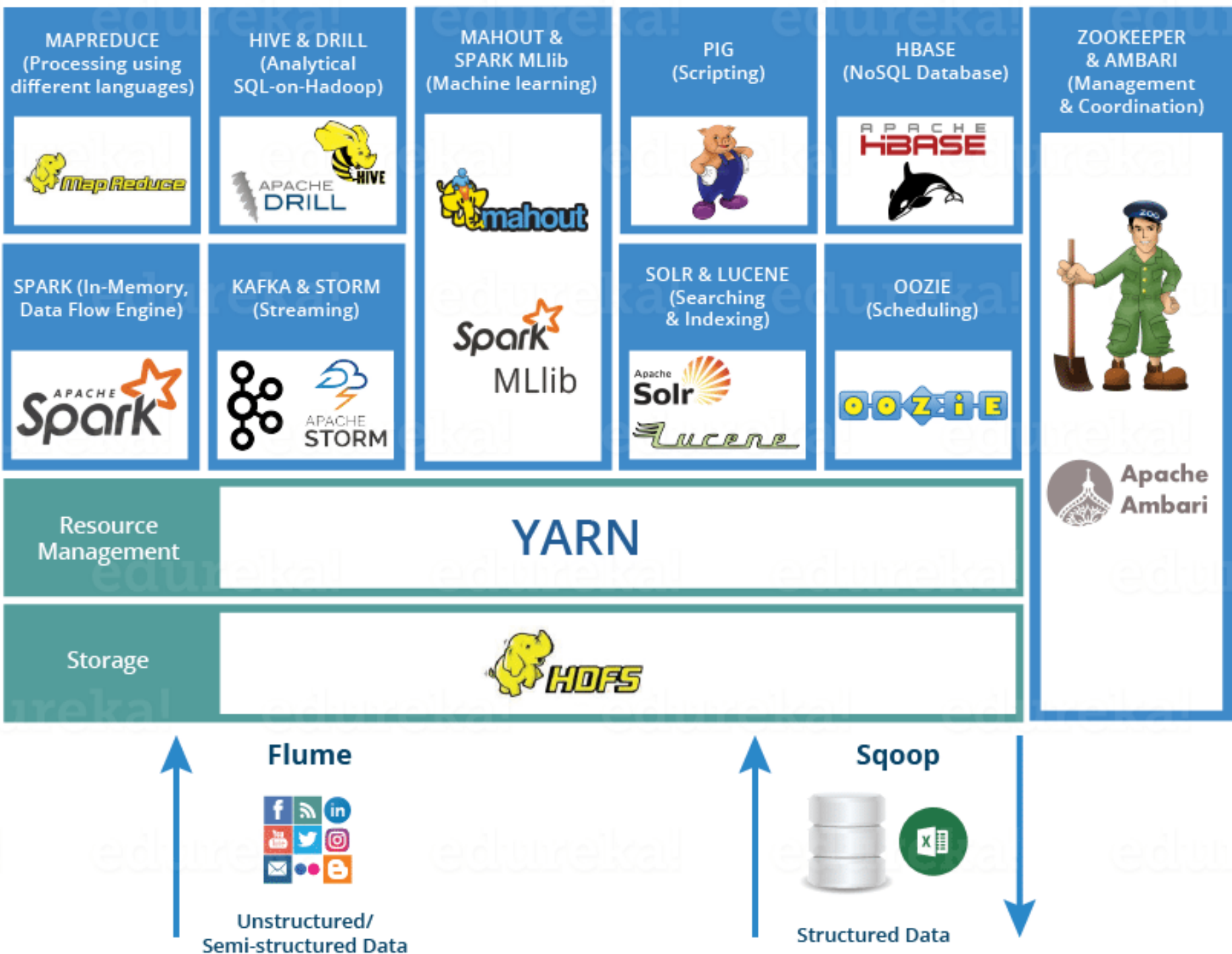
Vertical scaling: Pros and Cons

Advantages

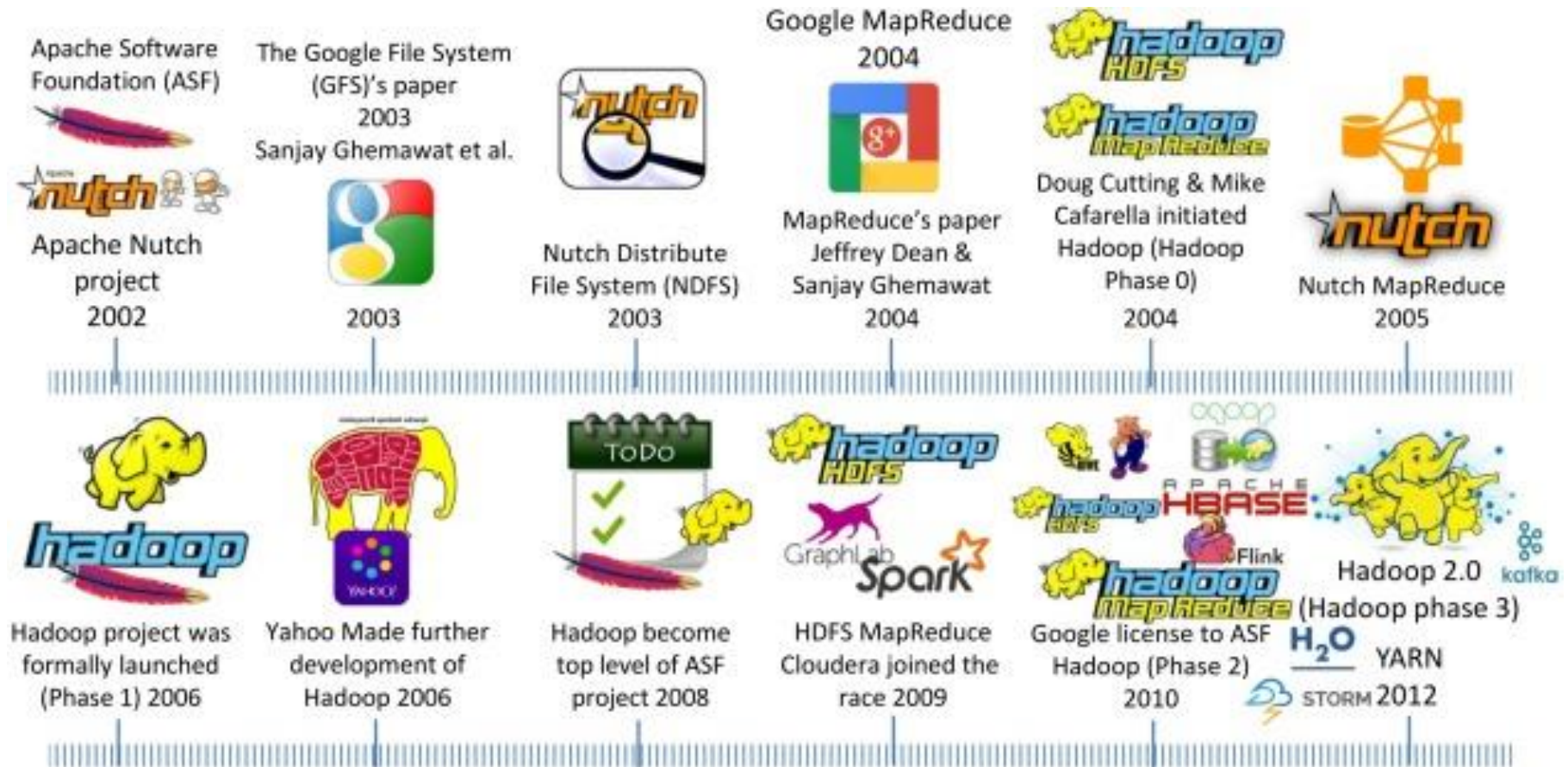
- Simpler to implement (no need for load balancing)
- Requires fewer changes to applications.

Disadvantages

- Limited by hardware constraints.
- More expensive as high-end hardware can be costly.
- If the machine fails, the entire system can be impacted.



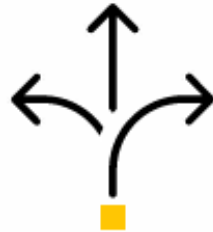
A brief history of Hadoop



Hadoop framework: Benefits



Fast



Flexible



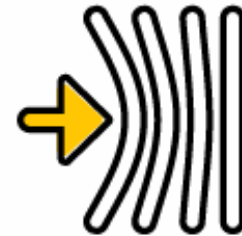
Scalable



Cost-effective



High throughput



Resilient to
failure

Hadoop framework: Benefits

- **Fast:** Hadoop makes data processing hassle-free and faster.
 - It has been found that Hadoop can process terabytes of unstructured data in just a few minutes, while petabytes in hours.
- **Flexible:** Hadoop helps gather data from different sources, of various types, and get valuable insights for many purposes.
 - Data sources: social media, emails, etc.
 - Data types: structured, semi-structured, or unstructured.
 - Purposes: log processing, market analysis, fraud detection, etc..
- **Scalable:** Businesses can store and distribute large data sets from hundreds of servers that operate parallelly.

Hadoop framework: Benefits

- **Cost-effective:** Hadoop runs on commodity hardware.
 - One can easily increase nodes without suffering from any downtime of pre-planning requirements.
- **High throughput:** More jobs can be done in less time.
 - A small job is split into multiple chunks of data in parallel, which are easier to handle.
- **Resilient:** It is possible to recover data whenever any node goes down.
 - It stores replicas of every block at different nodes in the cluster.

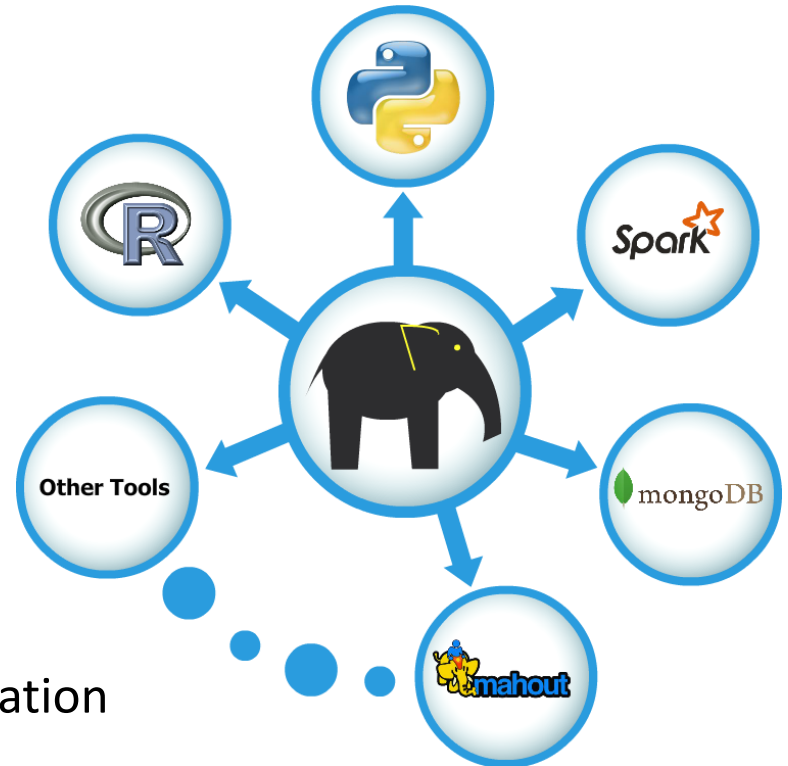
Strategic planning for scalable Hadoop

- You should first understand the level of complexity of data and the rate with which it is going to grow.
- Build a small or medium cluster for the data available at present and scale up the cluster in future.



Hadoop as a multi-purpose framework

- Hadoop can be integrated with multiple analytic tools to get the best out of it.



Mahout for Machine-Learning

Spark for real time processing

Pentaho for BI, etc.

R and **Python** for analytics and visualization

MongoDB and **Hbase** for NoSQL database

Hadoop framework: Limitations



Issues with
small files



Iterative processing



Low security



Higher vulnerability



Support only
Batch processing

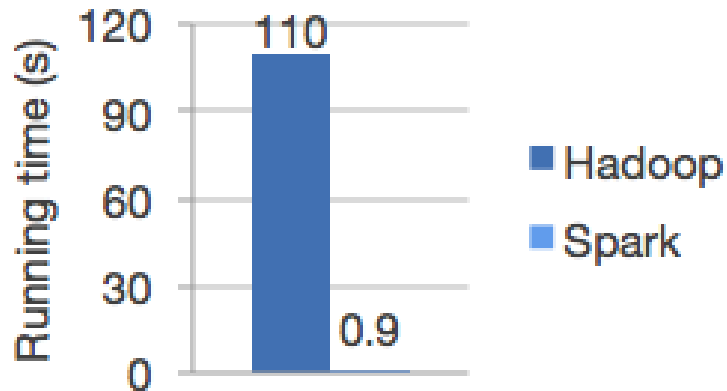
Hadoop framework: Limitations

- **Issues with small files:** Hadoop lacks the potential to support random reading of small files efficiently and effectively.
 - A small file is comparatively smaller than the HDFS block size.
 - A vast number of small files may overload the HDFS namespace.
- **Iterative processing:** Hadoop is an unfit choice for machine learning or iterative processing-based solutions.
 - The data flow in Hadoop framework is in the form of a chain, such that the output of one becomes the input of another stage.

Hadoop framework: Limitations

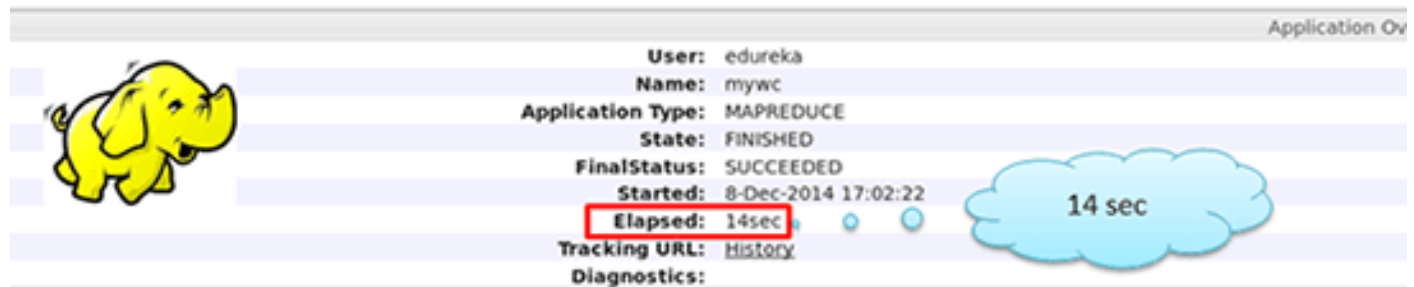
- **Low security:** Security model is disabled by default.
 - Hadoop does not offer encryption at the storage and network levels.
- **Higher vulnerability:** Cybercriminals may easily get access to Hadoop-based solutions.
 - Java is a popular, yet heavily exploited programming language.
- **Support only Batch processing:** MapReduce fails to take advantage of memory to the maximum.

Hadoop vs. Spark: Performance



Spark runs programs up to 100x faster than Hadoop MapReduce in memory, or 10x faster on disk.

A simple example of line processing in Hadoop and Spark



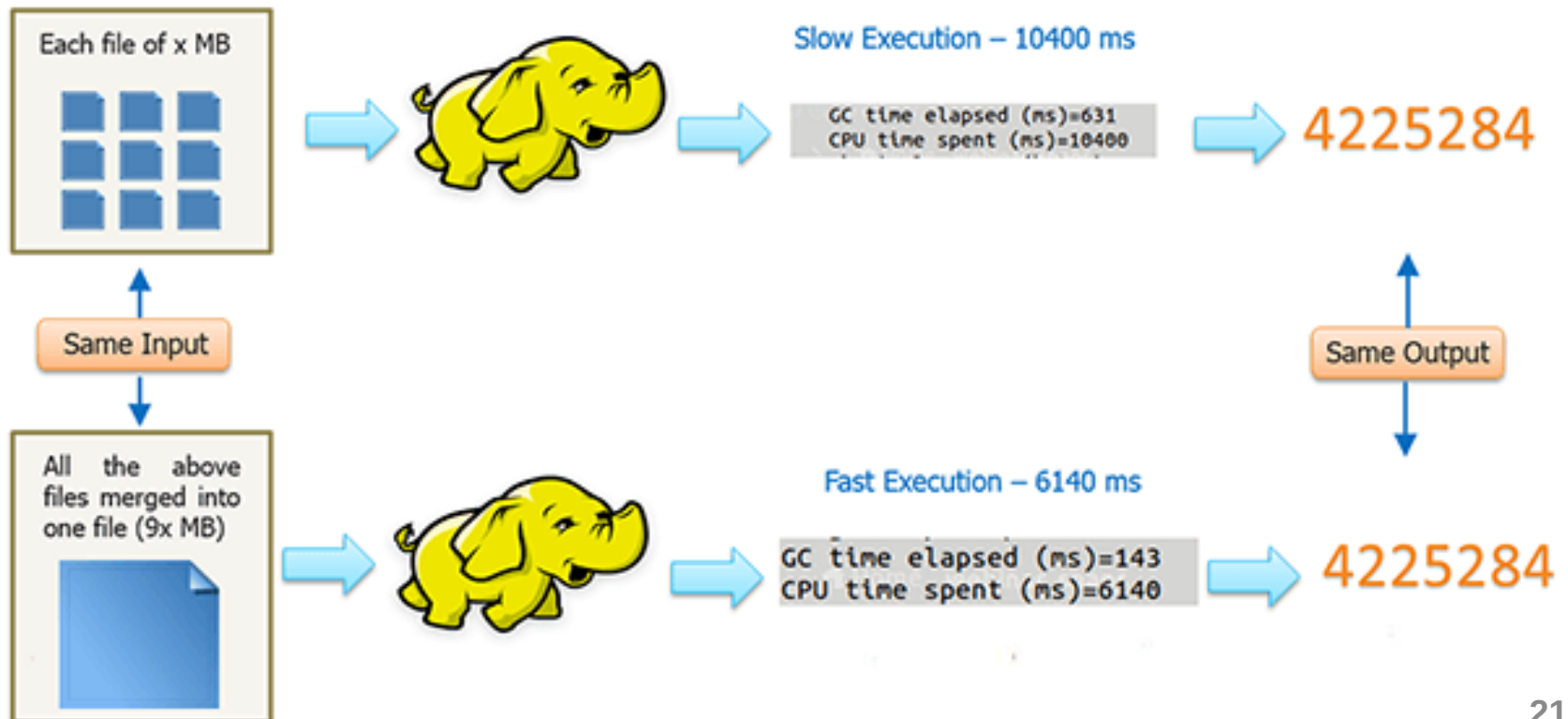
Logistic regression in Hadoop and Spark



```
14/12/08 04:10:06 INFO spark.SparkHadoopWriter: attempt_201412080410_0005_m_000000_5: Committed
14/12/08 04:10:06 INFO scheduler.TaskSetManager: Finished task 0.0 in stage 5.0 (TIO 5) in 296 ms on localhost (1/1)
14/12/08 04:10:06 INFO scheduler.TaskSchedulerImpl: Removed TaskSet 5.0, whose tasks have all completed, from pool
14/12/08 04:10:06 INFO scheduler.DAGScheduler: Stage 5 (saveAsTextFile at <console>:14) finished in 0.279 s
14/12/08 04:10:06 INFO spark.SparkContext: Job finished: saveAsTextFile at <console>:14, took 0.626043309 s
14/12/08 04:10:06 INFO executor.Executor: Finished task 0.0 in stage 5.0 (TIO 5). 826 bytes result sent to driver
wordCounts: Unit = ()
```

Hadoop with multiple small files

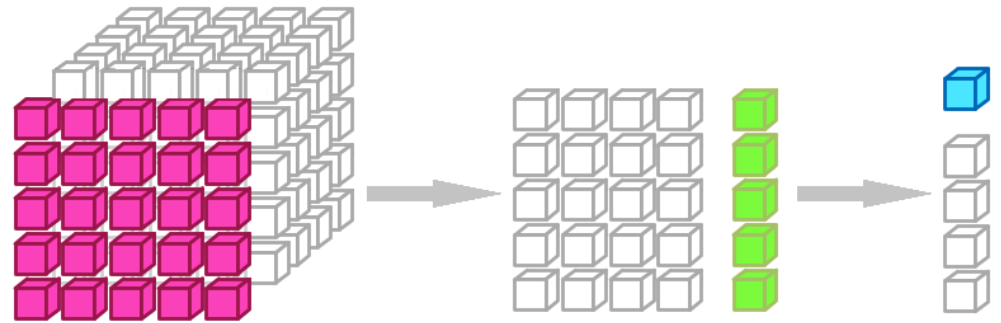
- We can merge the small files (e.g., server daily logs) into one big file and finally run our MapReduce program on it.
 - They are of the same format and handled by the same processing.



Hadoop with data processing

- Hadoop complements current RDBMS technologies, not a replacements.

Hadoop is not a good choice for OLTP or OLAP/DSS.



- It cannot perform well for intensive calculations with little data, which is common in deep learning.
- Different tools for different jobs, as simple as that.



Hadoop case studies

Hadoop in action

- There have several enterprises implementing Hadoop.



Hadoop case studies in enterprise

NETFLIX

Recommendation engines
and new content decisions
Hadoop, Hive and Pig
together with traditional BI

U.S. XPRESS

Collect and store 100s of data
points from thousands of
trucks, plus lots of geo data in
a Hadoop cluster



Product recommendations, in-store
shopping experience of customers
Hadoop and NoSQL technologies

More at the following [link](#)

Hadoop case studies in academy



About **200 million** text pages

Use Hadoop to distribute the workload for loading the information into memory



Recommender system of pins and boards using GCN

Hadoop2 cluster with 378 d2.8xlarge Amazon AWS nodes

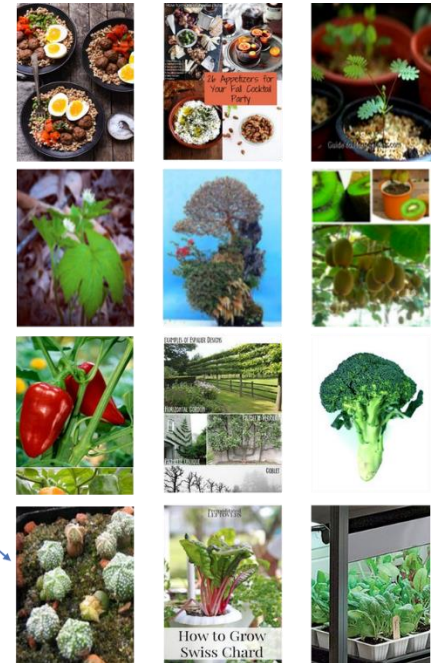


Visual

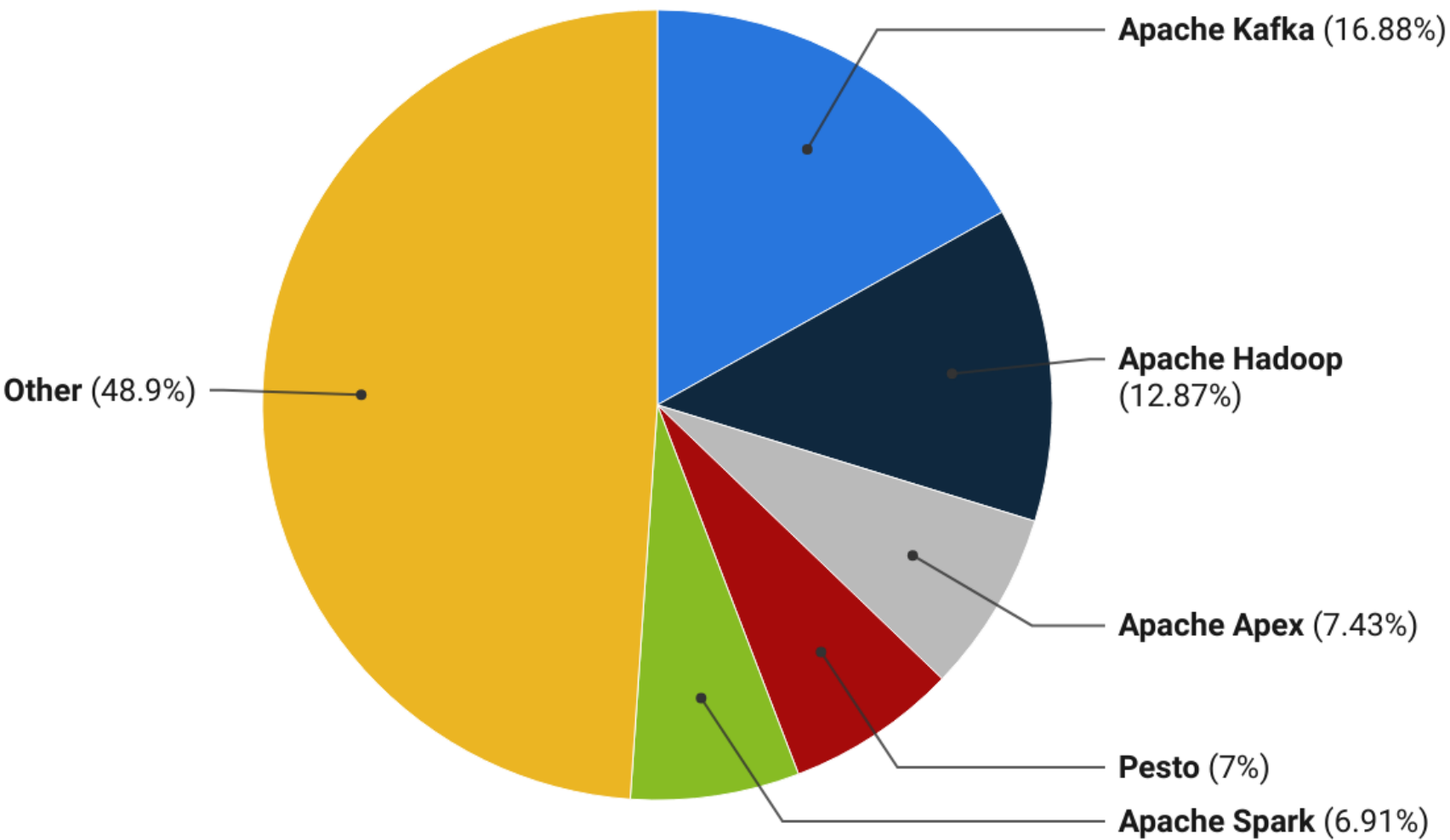
Annot.

Pixie

PinSage



Market Share Of Leading Big Data Processing Technologies Worldwide In 2023





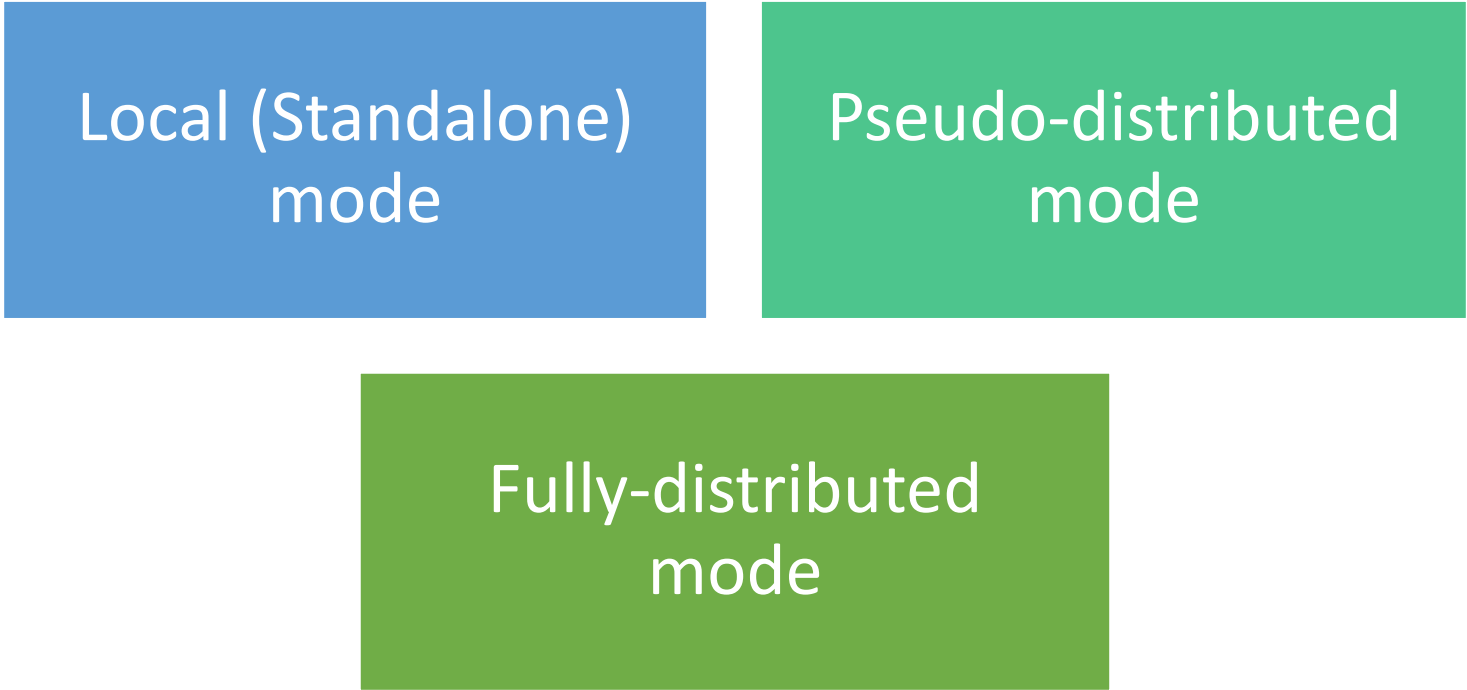
Hadoop installation

The installation process



Installation modes

- Hadoop cluster can be set up in one of the following three supported modes:



Local (Standalone)
mode

Pseudo-distributed
mode

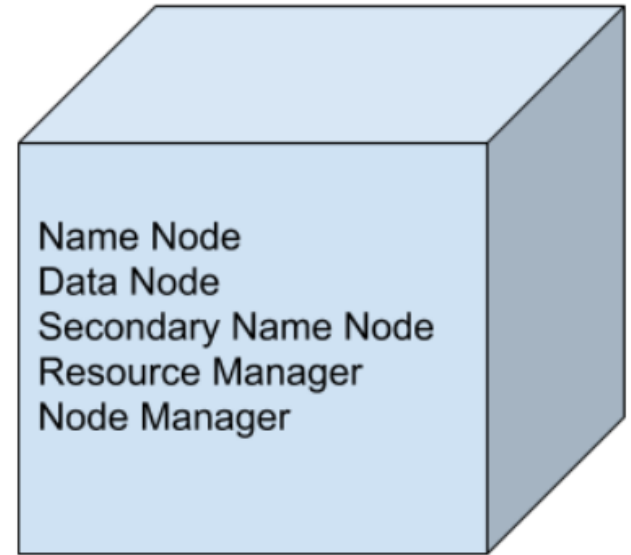
Fully-distributed
mode

Local (Standalone) mode

- Hadoop is **by default** configured to run **all the processes in a single JVM** (Java Virtual Machines).
- This is useful for learning, testing, and debugging.
- There is **no need to configure the xml files**—hdfs-site.xml, mapred-site.xml, core-site.xml for Hadoop environment.

Pseudo-distributed mode

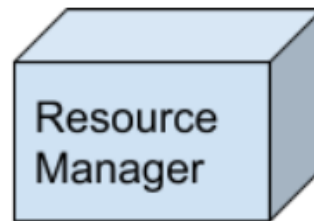
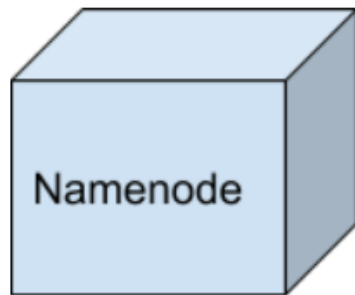
- Use only a single node to simulate the cluster.
- All the processes run independently on separate JVMs.



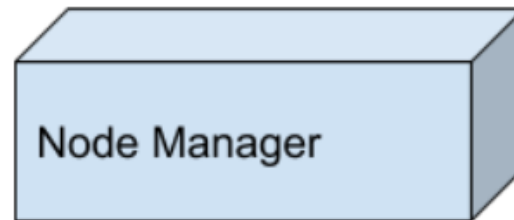
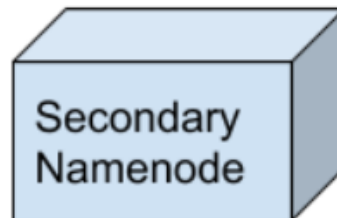
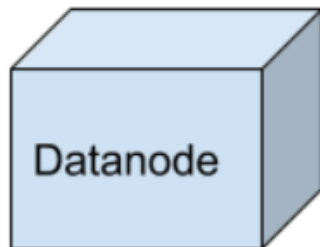
- This is both for development and debugging purposes.
- The configuration files must be specified properly.

Fully-distributed mode

- Hadoop runs on the cluster of machines, each of which plays the role of master daemon or slave daemons.
- The data is distributed across different nodes.



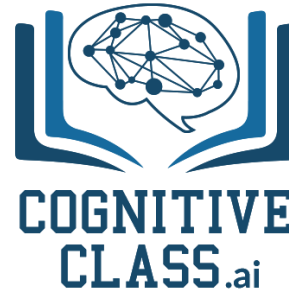
This is the production mode of Hadoop.



Top Hadoop Technology companies



Where to
learn
Hadoop?



edureka!



cloudera®
coursera

Alternatives to Hadoop



APACHE
STORM[™]



Apache Flink



Google
Big Query



amazon
EMR



databricks

CLOUDERA

...the end.

